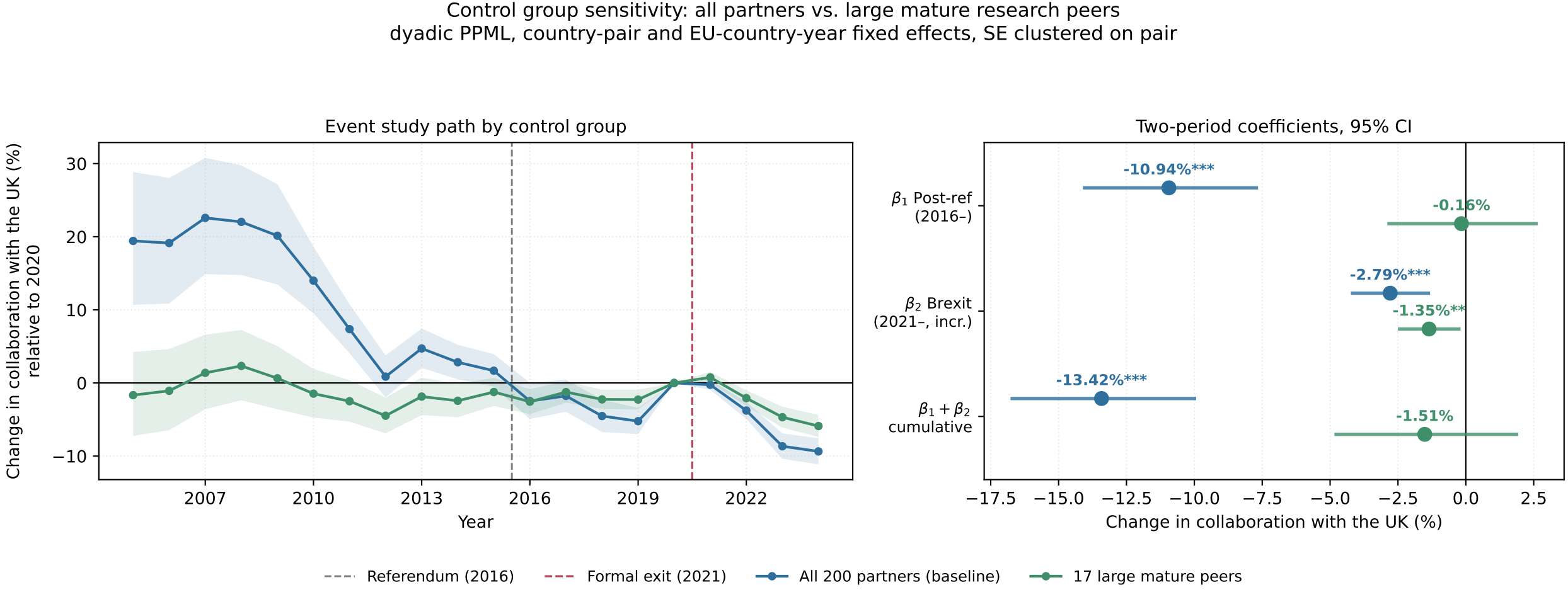
Control group sensitivity: all partners vs. large mature research peers
dyadic PPML, country-pair and EU-country-year fixed effects, SE clustered on pair
Event study path by control group
Change in collaboration with the UK (%) relative to 2020
30
20
10
0
−10
2007
2010
2013
2016
2019
2022
Year
Two-period coefficients, 95% CI
β_1 Post-ref (2016–)
β_2 Brexit (2021–, incr.)
$\beta_1 + \beta_2$ cumulative
-10.94%***
-0.16%
-2.79%***
-1.35%**
-13.42%***
-1.51%
−17.5
−15.0
−12.5
−10.0
−7.5
−5.0
−2.5
0.0
2.5
Change in collaboration with the UK (%)
Referendum (2016)
Formal exit (2021)
All 200 partners (baseline)
17 large mature peers